a

Human protein-coding genes
19,324

FIVE REFERENCE DATABASES — no LLM
5,625 flagged surface
UniProt · GO CC · HPA · SURFY · CSPA

ACCESSIBILITY TRIAGE AGENT — Sonnet
STAGE 1 NCBI gene + protein records
19,324 screened → 4,249 surface
STAGE 2 literature pass on the non-surface calls
14,404 non-surface calls re-read
12,987 with no database flag · 1,417 with one
→ 379 reclassified as surface
4,628 called surface
2,557 yes · 2,071 contextual
1,108 flagged by no database

TRIMMED FROM THE UNION
−1,401
one database flag, plus a high-confidence non-surface call (cytoplasmic, inner-leaflet, endomembrane, secreted, nuclear, mitochondrial)

PER-GENE DEEP DIVE
5,332

b

STAGE 1
Evidence retrieval
Surface Evidence + Surface Biology agents

DETERMINISTIC SEARCH SEEDS
Template-driven query anchors, fired in parallel (no LLM)

LITERATURE RETRIEVAL (no LLM)
PubMed · Europe PMC · PubTator

PER-PAPER PIPELINE
- **Abstract triage** — Discard · keep abstract · worth fetching — Haiku
- **Full-text fetch** — PMC; Unpaywall PDF fallback when PMC has no body — Tool
- **Trim** — Per-paper clip pruning — Haiku
- **Select** — Final evidence picks — Sonnet

STAGE 2
Block builders
Run in parallel; one Sonnet call per block
Sonnet

SURFACE EVIDENCE
- Evidence quality
- Detection methods
- Conflicting evidence

SHARED
- Risks

SURFACE BIOLOGY
- Evidence quality
- Tissue expression
- Subcellular location
- Anatomical accessibility
- Accessibility modulation

STAGE 3
Synthesizer
Sonnet

Reads
- Block drafts
- Evidence ledgers (STAGE 1)
- Structural + topology data
  DeepTMHMM · AlphaFold · PDBe SIFTS
  SURFACE-Bind · Compara orthologs
  Schweke 2024 homo-oligomers
- Triage prior

Emits
- Executive summary
- Filters
- Confidence + rationale

Sonnet claude-sonnet-4-6 · Haiku claude-haiku-4-5 · Tool deterministic / retrieval (no LLM)